13 Consolidation, starter routines and transition to Year 12 (Première)

Sheet 75 Tue 1st June 2027 *Synthesis problems drawing on several topics*

1. Warm-up · Quick Solve $3(x-1) = x+5$.
2. Core · Method For which value of k are the points $A(1,1)$, $B(3,5)$ and $C(k,9)$ collinear?
3. Core + · Method A price increases by 10% then decreases by 10%. What is the overall change?
4. Review · Quick Total surface area of a cuboid with dimensions 4, 3, 2 (cm)?
5. Thread · Quick Calculate the exact volume of a sphere of radius 3 cm.

Sheet 76 Thu 3 June 2027 *Review of the six competences*

1. Warm-up · Metacognition List the six mathematical competences developed at lycée.
2. Core · Metacognition Match each of the following to a competence: "translating a problem statement into an equation"; "sketching a curve"; "justifying an equality with a written calculation".
3. Core + · Metacognition To solve $2x+3=11$, which competence is mainly used?
4. Review · Problem Water at 20 °C is heated by 3 °C per minute: $T(t) = 20 + 3t$. After how many minutes does it reach 65 °C?
5. Thread · Quick Solve in $\mathbb{R}$: $0.5x + 1.2 = 2.7$.

Sheet 77 Mon 7 June 2027 *Targeted reactivation of prerequisites for Première*

1. Warm-up · Quick Expand $(x+3)^2$.
2. Core · Method Check that $x^2 - 5x + 6 = (x-2)(x-3)$ and deduce the solutions of $x^2 - 5x + 6 = 0$.
3. Core + · Anticipation For $f(x) = x^2$, calculate the rate of change $\dfrac{f(3)-f(1)}{3-1}$.
4. Review · Construction Place M and N, the midpoints of $[AB]$ and $[AC]$, then draw $[MN]$. What can be said about (MN) and (BC)? Calculate MN if $BC = 9$ cm.
5. Thread · Write-up Prove that a triangle with sides 7, 24 and 25 is right-angled.

Sheet 78 Tue 8 June 2027 *Synthesis problems: choosing and justifying a strategy*

1. Warm-up · Reading code What does $f(2)$ return?

```
def f(x):
    return x**2 - 3*x + 1
```

2. Core · Completion Complete the code so that the function returns `True` when A, B, C are collinear:

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == ....
```

3. Core + · Completion Complete the absolute value function:

```
def vabs(x):
    if x >= 0:
        return x
    else:
        return ....
```

4. Review · Quick Write 2.5×10^3 and then 6.2×10^{-2} in decimal notation.
5. Thread · Quick Give the reflection of $A(2,-3)$ in the origin.

Sheet 79 Thu 10 June 2027 *Presentation of the specialist subjects; review*

1. Warm-up · Quick Express $\frac{3}{8}$ as a percentage.
2. Core · Metacognition What are your three revision priorities before Première?
3. Core + · Quick Solve $2x^2 = 18$.
4. Review · Trap Calculate $(-2)^4$ and -2^4.
5. Thread · True/false True or false? Justify. If an integer is divisible by 4 and by 6, then it is divisible by 24.